

Figure 9 plots the mean and variance of the KDE for different bandwidths.

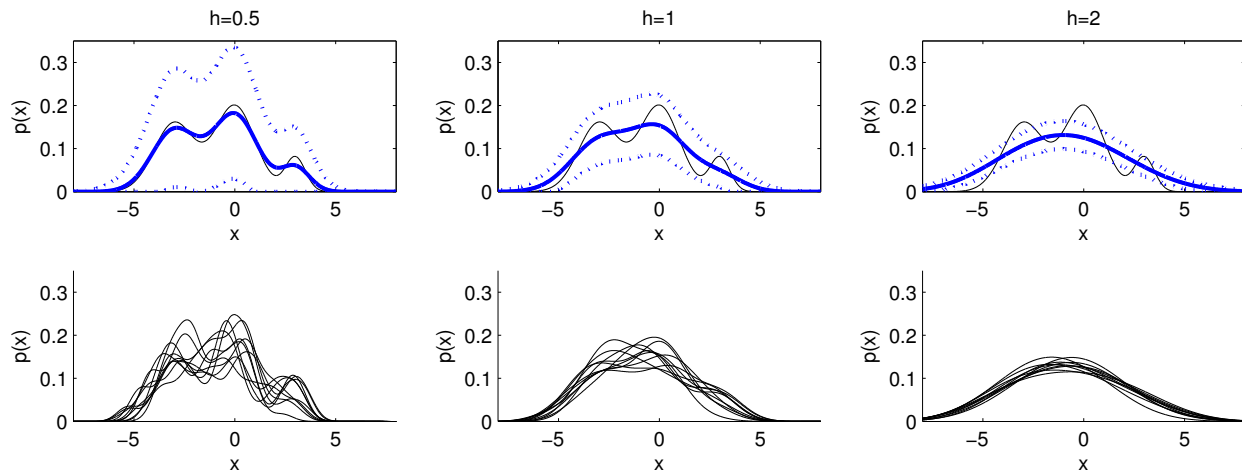


Figure 9: Mean and variance of kernel density estimator. Top row: the true density function (thin line) and the estimator mean (thick line) and 2 standard deviations around the mean (dotted line) for different bandwidths h . Bottom row: Examples of the estimate $\hat{p}(x)$ using 10 different sets of 50 samples drawn from the true density $p(x)$. When the bandwidth is small ($h = 0.5$), the estimates are significantly different for each sample set (i.e., high variance), but the estimator mean is close to the true density. When the bandwidth is large ($h = 2$), the estimates are consistent with each other (i.e., low variance), but the estimator mean is far from the true density.